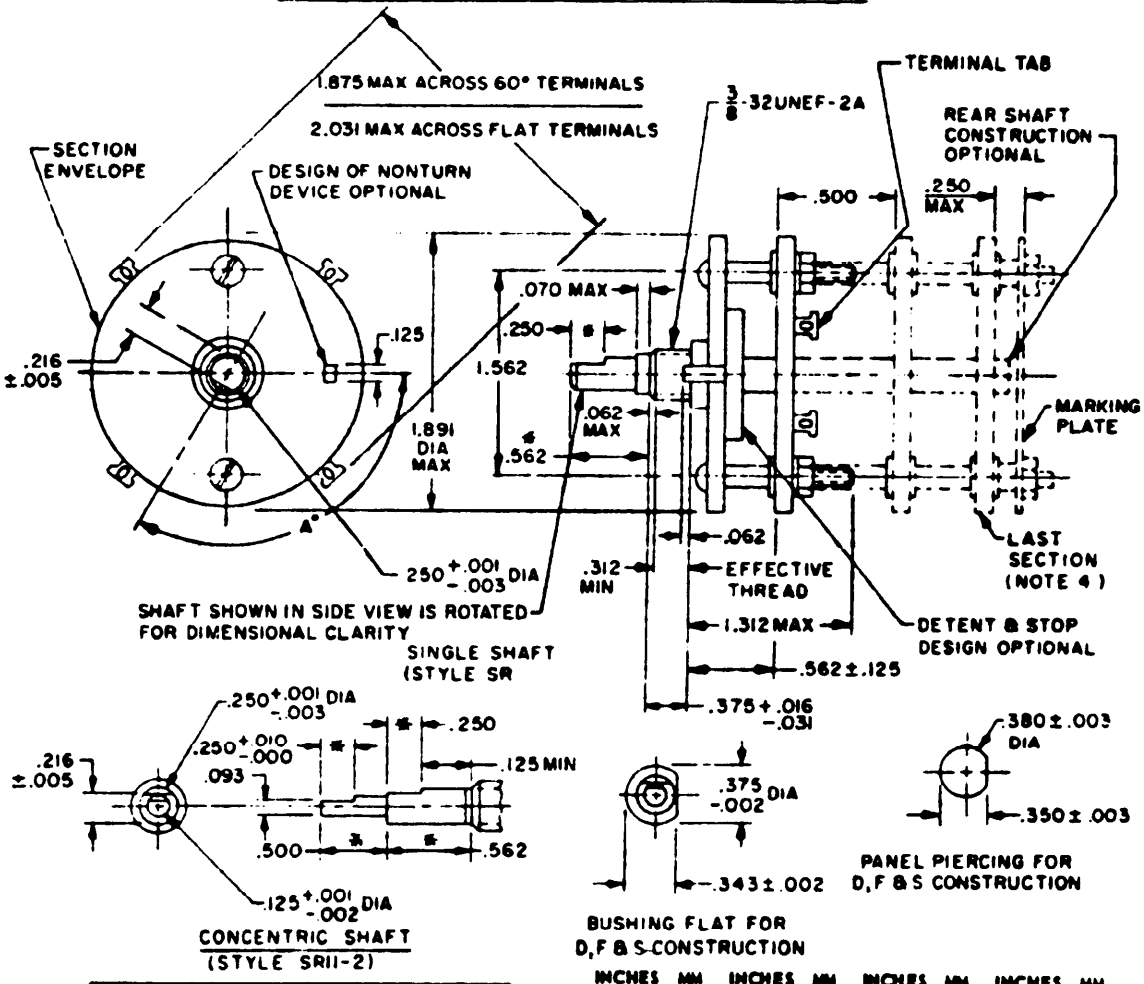
MIL-S-3786/11C
18 December 1968
SUPERSEDING
MIL-S-3786/11B
5 November 1965

MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, OPEN CONSTRUCTION, 1/2 AMPERE
STYLE SR11

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-3786.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION SHEET

USE MIL-S-3786/1

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.062	1.57	.312	7.92	.562	14.27
.002	.05	.070	1.78	.343	8.71	1.312	33.32
.003	.08	.093	2.36	.350	8.89	1.562	39.67
.005	.13	.125	3.18	.375	9.53	1.875	47.63
.010	.25	.216	5.49	.380	9.65	1.891	48.03
.016	.41	.250	6.36	.500	12.70	2.031	51.59
.031	.79						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm).
4. The number of sections is optional.
5. Nonturn device as shown is not applicable to S construction switches.
6. Shaft-flat angle A° is the angle between a line through the center of the shaft and center of the nonturn device and another line through the center of the shaft and perpendicular to the shaft flat.
7. Shaft shown in maximum counterclockwise position for switches with stop and with switch in position No. 1 for switches without stop.
8. For allowable variations, see MIL-S-3786 ordering data.
9. Front plate design optional provided it falls within the maximum O.D. of the section dimension referenced.

FSC 5930

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REQUIREMENTS:

Dimensions and configuration: See figure.

Angle of throw: 18°, 20°, and 30°.

Terminals: The terminal tabs shall be bent at an angle of 60 ± 15 degrees from the plane of the section. When flat terminals are required, the dimensions shall be as shown on figure.

Rotational torque: The minimum and maximum values of torque determined for shaft rotation shall be within the limits specified in table I.

Enclosure design: N or S.

TABLE I. Rotational-torque limits.

Temperature	Torque (lb-in)	
	Minimum	Maximum
Room	2	6
Minimum	2	8

Life (rotational): The test loads for the applicable circuit conditions shall be as specified in table II. Each of the loads, specified for the applicable environmental condition, shall be switched by at least one rotor contact of the switch.

TABLE II. Electrical loads.

Environmental condition	Inductive load (2.8 Henries)		Resistive loads	
	Milliamperes	Volts, dc	Milliamperes	Volts
At atmospheric pressure	50	28	500	28 dc
			250	115 vrms
			50	300 vrms
At reduced barometric pressure	---	---	350	28 dc
			100	115 vrms

Dielectric withstanding voltage: The applicable test voltage specified in table III shall be applied between the general switch elements.

TABLE III. Dielectric-test voltages.

Altitude	Test voltage (Volts, rms)
At atmospheric pressure	1,000
At reduced barometric pressure	400

Contact resistance: For switches of life-temperature characteristics A, C, and E, the contact resistance at room temperature shall not exceed the following values:

Initial and after vibration and shock - - - - 20 milliohms.
After moisture-resistance and salt spray - - - - 30 milliohms.
After initial soak and completion of life test - - - - 40 milliohms.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5930-0606-8)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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